

“Instituto Politécnico Nacional”

“Unidad Profesional Interdisciplinaria en
Ingeniería y Tecnologías Avanzadas”



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Grupo: 4TM3.

Escuela: Unidad Profesional
Interdisciplinaria en
Ingeniería y Tecnologías
Avanzadas.

Materia: Aplicaciones Distribuidas.

Profesor: Sierra Romero Noé.

Trabajo: Practica 06.

Practica 6 MongoDB

Práctica sobre la recuperación de registros de la Base de datos de AWS en mongo DB:

Usuario: kdocdany_db_user

Contraseña: IBIUz597KFx5jQwG

npm install mongodb

mongodb+srv://kdocdany_db_user:IBIUz597KFx5jQwG@cluster0.u4w7qod.mongodb.net/?appName=Cluster0

```
//Primera linea agregada para cambiar de servidores dns a los de google
```

```
require("node:dns/promises").setServers(["1.1.1.1", "8.8.8.8"]);
```

```
const MongoClient = require('mongodb').MongoClient;
```

```
const assert = require('assert');
```

```
function iterateFunc(doc) {
```

```
    console.log(JSON.stringify(doc, null, 4));
```

```
}
```

```
async function listDatabases(client) {
```

```
    databasesList = await client.db().admin().listDatabases();
```

```
    console.log("Databases:");
```

```
    databasesList.databases.forEach(db => console.log(` - ${db.name}`));
```

```
};
```

```
async function findAllData(client) {
```

```
const cursor = await client.db("sample_mflix").collection("movies").find({}).limit(2);

// Convertir cursor a array de documentos

const results = await cursor.toArray();

console.log("Title: ",results[0]["title"]);


// Mostrar resultados

console.log("Películas encontradas:");

console.log(JSON.stringify(results, null, 2));

}

async function main() {

const uri =
"mongodb+srv://kdocdany_db_user:IBIUz597KFx5jQwG@cluster0.u4w7qod.mongodb.net/?appName=
Cluster0";

const client = new MongoClient(uri, {

  family: 4,

  connectTimeoutMS: 10000

});

try {

  // Connect to the MongoDB cluster

  await client.connect();


  // Make the appropriate DB calls

  await listDatabases(client);

  await findAllData(client);

}
```

```

    } catch (e) {

        console.error(e);

    } finally {

        await client.close();

    }

}

main().catch(console.error);

```

```

Administrador: Símbolo del sistema
Microsoft Windows [Versión 10.0.22631.6190]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Windows\System32>cd D:\Documents\UNI papeles\11º Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_ServerMongoDB

C:\Windows\System32>D:

D:\Documents\UNI papeles\11º Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_ServerMongoDB>npm init -d
npm info using npm@11.6.2
npm info using node@24.13.0
This utility will walk you through creating a package.json file.
It only covers the most common items, and tries to guess sensible defaults.

See 'npm help init' for definitive documentation on these fields
and exactly what they do.

Use 'npm install <pkg>' afterwards to install a package and
save it as a dependency in the package.json file.

Press ^C at any time to quit.
package name: (proyecto6_servermongodb)
version: (1.0.0)
description: Extrayendo registros de la base de datos de mongoDB
entry point: (server.js)
test command:
git repository:
keywords:
author: Daniel OC
license: (ISC)
type: (commonjs)
About to write to D:\Documents\UNI papeles\11º Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_ServerMongoDB\package.json:

{
  "name": "proyecto6_servermongodb",
  "version": "1.0.0",
  "description": "Extrayendo registros de la base de datos de mongoDB",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "author": "Daniel OC",
  "license": "ISC",
  "type": "commonjs"
}

Is this OK? (yes) yes

```

Is this OK? (yes) yes

npm info ok

```

D:\Documents\UNI papeles\11º Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_ServerMongoDB>npm install mongod
added 12 packages, and audited 13 packages in 3s

found 0 vulnerabilities

```

```
Administrador: Símbolo del sistema
D:\Documents\UNI papeles\11° Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_Server\MongoDB>node server.js
Databases:
- sample_mflix
- admin
- local
Title: In the Land of the Head Hunters
Películas encontradas:
[
  {
    "id": "573a1390f29313caabcd516c",
    "plot": "Original advertising for the film describes it as a drama of primitive life on the shores of the North Pacific...",
    "genres": [
      "Drama",
      "History"
    ],
    "runtime": 65,
    "cast": [
      "Stanley Hunt",
      "Sarah Constance Smith Hunt",
      "Mrs. George Walkus",
      "Paddy 'Malid'"
    ],
    "num_mflix_comments": 1,
    "poster": "https://m.media-amazon.com/images/M/MW5BMjE3MjA5NzNSIN5BM15BanBnXkFtZTgwMjASOTg5NjE@._V1_SY1000_SX677_AL_.jpg",
    "title": "In the Land of the Head Hunters",
    "lastupdated": "2015-09-16 12:11:37.770000000",
    "languages": [
      "English"
    ],
    "released": "1914-12-07T00:00:00.000Z",
    "directors": [
      "Edward S. Curtis"
    ],
    "writers": [
      "Edward S. Curtis (story)"
    ],
    "awards": {
      "wins": 1,
      "nominations": 0,
      "text": "1 win."
    },
    "year": 1914,
    "imdb": {
      "rating": 5.8,
      "votes": 223,
      "id": 4150
    },
    "countries": [

```

```
Administrador: Símbolo del sistema
    "countries": [
      "USA"
    ],
    "type": "movie",
    "tomatoes": {
      "viewer": {
        "rating": 2.7,
        "numReviews": 64,
        "meter": 18
      },
      "dvd": "2000-08-15T00:00:00.000Z",
      "website": "http://www.milestonefilms.com/",
      "production": "World Film Corporation",
      "lastUpdated": "2015-08-18T19:21:12.000Z"
    },
    {
      "id": "573a1390f29313caabcd587d",
      "plot": "At 10 years old, Owens becomes a ragged orphan when his sainted mother dies. The Conways, who are next door neighbors, take Owen in, but the constant drinking by Jim soon puts Owen on the ...",
      "genres": [
        "Biography",
        "Crime",
        "Drama"
      ],
      "runtime": 72,
      "rated": "PASSED",
      "cast": [
        "John McCann",
        "James A. Marcus",
        "Maggie Weston",
        "Harry McCoy"
      ],
      "num_mflix_comments": 1,
      "poster": "https://w.media-amazon.com/images/M/MW5BN0kxZGU4MmMtOD11Ny00YzAzLTgAZGh0NGF1NzA5NzcxOTM1XkEyXkFqcGdeQXVyOTM3Mjc5MjI@._V1_SY1000_SX677_AL_.jpg",
      "title": "Regeneration",
      "fullplot": "At 10 years old, Owens becomes a ragged orphan when his sainted mother dies. The Conways, who are next door neighbors, take Owen in, but the constant drinking by Jim soon puts Owen on the street. By 17, Owen learns that might is right. By 25, Owen is the leader of his own gang who spend most of their time gambling and drinking. But Marie comes into the gangster area of town and everything changes for Owen as he falls for Marie. But he cannot tell her so, so he comes to her settlement to find education and inspiration. But soon, his old way of life will rise to confront him again.",
      "languages": [
        "English"
      ],
      "released": "1915-09-13T00:00:00.000Z",
      "directors": [
        "Raoul Walsh"
      ],
      "writers": [
        "Owen Frawley Kildare (book)",
        "Raoul Walsh (adapted from the book: 'My Mamie Rose')",
        "Carl Harbaugh (adapted from the book: 'My Mamie Rose')"
```

```
Administrador: Símbolo del sistema
"Carl Hiarbaugh (adapted from the book: \"My Mamie Rose\")"
},
"awards": {
  "wins": 1,
  "nominations": 0,
  "text": "1 win."
},
"lastupdated": "2015-08-14 01:28:18.957000000",
"year": 1915,
"imdb": {
  "rating": 6.8,
  "votes": 626,
  "id": 5960
},
"countries": [
  "USA"
],
"type": "movie",
"tomatoes": {
  "viewer": {
    "rating": 3.4,
    "numReviews": 395,
    "meter": 70
  },
  "critic": {
    "rating": 9.2,
    "numReviews": 5,
    "meter": 100
  },
  "lastUpdated": "2015-09-17T17:22:42.000Z",
  "rotten": 0,
  "production": "Fox Film Corporation",
  "fresh": 5
}
}
]
D:\Documents\UNI papeles\11º Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_ServerMongoDB>
```

Ejemplos adicionales con comments y con embedde_movies y se van a recuperar los primeros 5 registros

//Primera linea agregada para cambiar de servidores dns a los de google

```
require("node:dns/promises").setServers(["1.1.1.1", "8.8.8.8"]);
```

```
const MongoClient = require('mongodb').MongoClient;
```

```
const assert = require('assert');
```

```
function iterateFunc(doc) {
```

```
  console.log(JSON.stringify(doc, null, 4));
```

```
}
```

```
async function listDatabases(client) {
```

```
  databasesList = await client.db().admin().listDatabases();
```

```
  console.log("Databases:");
```

```
databasesList.databases.forEach(db => console.log(` - ${db.name}`));

};

async function findAllData(client) {

  const cursor = await client.db("sample_mflix").collection("movies").find({}).limit(2);

  // Convertir cursor a array de documentos

  const results = await cursor.toArray();

  console.log("Title: ",results[0]['title']);


  // Mostrar resultados

  console.log("Películas encontradas:");

  console.log(JSON.stringify(results, null, 2));

}

async function fetchFiveRecords(client, collectionName) {

  console.log(`\nExtrayendo 5 registros de: ${collectionName}`);

  const collection = client.db("sample_mflix").collection(collectionName).find({}).limit(5);

  const results = await collection.toArray();

  // Mostrar resultados

  console.log("Películas encontradas:");

  console.log(JSON.stringify(results, null, 2));

}

async function main() {
```

```
const uri =
"mongodb+srv://kdocdany_db_user:IBIUz597KFx5jQwG@cluster0.u4w7qod.mongodb.net/?appName=
Cluster0";

const client = new MongoClient(uri, {

  family: 4,

  connectTimeoutMS: 10000

});

try {

  // Connect to the MongoDB cluster

  await client.connect();

  // Make the appropriate DB calls

  await listDatabases(client);

  await findAllData(client);

  await fetchFiveRecords(client, "embedded_movies");

  await fetchFiveRecords(client, "comments");

} catch (e) {

  console.error(e);

} finally {

  await client.close();

}

}

main().catch(console.error);
```

```
Administrador: Símbolo del sistema
Extrayendo 5 registros de: embedded_movies
Películas encontradas:
[
  {
    "id": "573a1390f29313caabdc5293",
    "plot": "Young Pauline is left a lot of money when her wealthy uncle dies. However, her uncle's secretary has been named as her guardian until she marries, at which time she will officially take ...",
    "genres": [
      "Action"
    ],
    "runtime": 199,
    "cast": [
      "Pearl White",
      "Crane Wilbur",
      "Paul Panzer",
      "Edward Jose"
    ],
    "num_mflix_comments": 0,
    "poster": "https://m.media-amazon.com/images/H/W/5B/tzgX0Ok1Mzk2M15BM15BanBnXkFtZTgwODg0NzkwMjE@._V1_SV1000_SX677_AL_.jpg",
    "title": "The Perils of Pauline",
    "fullplot": "Young Pauline is left a lot of money when her wealthy uncle dies. However, her uncle's secretary has been named as her guardian until she marries, at which time she will officially take possession of her inheritance. Meanwhile, her \"guardian\" and his confederates constantly come up with schemes to get rid of Pauline so that he can get his hands on the money himself.",
    "languages": [
      "English"
    ],
    "released": "1914-03-23T00:00:00.000Z",
    "directors": [
      "Louis J. Gasnier",
      "Donald MacKenzie"
    ],
    "writers": [
      "Charles W. Goddard (screenplay)",
      "Basil Dickey (screenplay)",
      "Charles W. Goddard (novel)",
      "George B. Seitz",
      "Bertram Millhauser"
    ],
    "awards": {
      "wins": 1,
      "nominations": 0,
      "text": "1 win."
    },
    "lastupdated": "2015-09-12 00:01:18.647000000",
    "year": 1914,
    "imdb": {
      "rating": 7.6,
      "votes": 744,
      "id": 4465
    },
    "countries": [

```

```
Administrador: Símbolo del sistema
Extrayendo 5 registros de: comments
Películas encontradas:
[
  {
    "id": "5a9427648b0beebeb6957a88",
    "name": "Thomas Morris",
    "email": "thomas_morris@fakegmail.com",
    "movie_id": "573a1390f29313caabdc680a",
    "text": "Perspiciatis sequi nesciunt maiores. Molestiae earum odio voluptas animi ipsam. Dolorem libero temporibus omnis quaerat deleniti atque. Tempore delectus esse explicabo nemo.",
    "date": "2004-02-26T06:33:03.000Z"
  },
  {
    "id": "5a9427648b0beebeb6957abd",
    "name": "John Bishop",
    "email": "john_bishop@fakegmail.com",
    "movie_id": "573a1391f29313caabdc6f9b",
    "text": "Accusamus qui distinctio ut ab saepe tenetur. Quae optio aut eius deleniti veritatis error. Eligendi ducimus rerum recusandae doloribus. Natus quisquam expedita voluptatum voluptatibus natus quidem.",
    "date": "1972-04-16T14:52:53.000Z"
  },
  {
    "id": "5a9427648b0beebeb69579f5",
    "name": "John Bishop",
    "email": "john_bishop@fakegmail.com",
    "movie_id": "573a1390f29313caabdc446f",
    "text": "Id error ab at molestias dolorum incidunt. Non deserunt praesentium dolore nihil. Optio tempora vel ut quas. Vnus dicta numquam quasi. Rem totam cumque at eum. Ullam hic ut ea magni.",
    "date": "1975-01-21T00:31:22.000Z"
  },
  {
    "id": "5a9427648b0beebeb6957bdc",
    "name": "Jordan Medina",
    "email": "jordan_medina@fakegmail.com",
    "movie_id": "573a1391f29313caabdc8979",
    "text": "Adipisci vel dolores tenetur sit inventore. Doloribus dolor nesciunt voluptas saepe veritatis. Mollitia eum iure ut nam.",
    "date": "2011-11-21T08:57:05.000Z"
  },
  {
    "id": "5a9427648b0beebeb6957bd4",
    "name": "Alliser Thorne",
    "email": "owen_teale@gameofthron.es",
    "movie_id": "573a1391f29313caabdc7c9e",
    "text": "A ut dolor illum deleniti repellendus. Iste fugit in quas minus nobis sunt rem. Animi possimus dolor alias natus consequatur saepe. Nihil quam magni aspernatur nisi.",
    "date": "1982-07-06T15:10:41.000Z"
  }
]
D:\Documents\UNI papeles\11º Semestre\Aplicaciones Distribuidas\Proyectos\Proyecto6_Server\MongoDB>
```